ITSimplera Institute
AI/ML Internship Program

TASK DOCUMENTATION

Week 1 — Data Analysis & Business Understanding

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Contents

Exploratory Data Analysis on the Online Retail II Dataset

1. Project Overview

A retail company wants to understand customer purchase behavior before investing in a recommendation system. Acting as a data analyst, the objective of this task was to explore the Online Retail II dataset, identify patterns and data quality issues, and produce a complete Exploratory Data Analysis (EDA) report covering who the customers are, which products sell the most, and which countries generate the most revenue.

1.1 Dataset

- Name: Online Retail II
- Source: UCI Machine Learning Repository (archive.ics.uci.edu/ml/datasets/Online+Retail+II)
- Description: Real transaction data from a UK-based online retailer, covering December 2009 to December 2011, including invoices, product descriptions, quantities, prices, and customer IDs.
- Format: Excel workbook with two sheets "Year 2009-2010" and "Year 2010-2011" combined for the full analysis period.

1.2 Tools Used

- Python 3 (Google Colab / Jupyter Notebook)
- pandas, numpy — data loading and manipulation
- matplotlib, seaborn — data visualization

1.3 Approach

Per the task requirements, this analysis is strictly observation-only no rows were removed or modified at any stage. All data quality issues (missing values, duplicates, cancellations, outliers) are identified and reported, not cleaned, so the company gets a full, honest picture of the raw data before any preprocessing decisions are made.

2. Step-by-Step Implementation

Step 1: Import Required Libraries

The core Python libraries for data handling and visualization were imported, and a consistent visual theme was set for all charts.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid")
```

Step 2: Load the Dataset

The Excel file contains two sheets covering 2009-2010 and 2010-2011. Both sheets were read separately and concatenated into a single DataFrame, so the analysis covers the full two-year transaction history rather than only one sheet.

```
sheet_2009_2010 = pd.read_excel(file_path, sheet_name="Year 2009-2010")
sheet_2010_2011 = pd.read_excel(file_path, sheet_name="Year 2010-2011")
df = pd.concat([sheet_2009_2010, sheet_2010_2011], ignore_index=True)
```

Basic dataset information was then displayed — shape, column names, and data types.

```
Dataset Shape: Rows: 1,067,371, Columns: 8
Columns: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country
Date Range: 2009-12-01  to  2011-12-09
```

Step 3: Check Missing Values and Duplicate Rows

Missing values and duplicate rows were identified and quantified, along with other data quality flags (negative quantities, negative or zero prices) that are relevant for a retail dataset.

```
Missing Values:
Description: 4,382 (0.41%)
Customer ID: 243,007 (22.77%)

Duplicate Rows: 34,335

Negative Quantity (returns/cancellations): 22,950
Zero Price rows: 6,202
Negative Price rows: 5
```

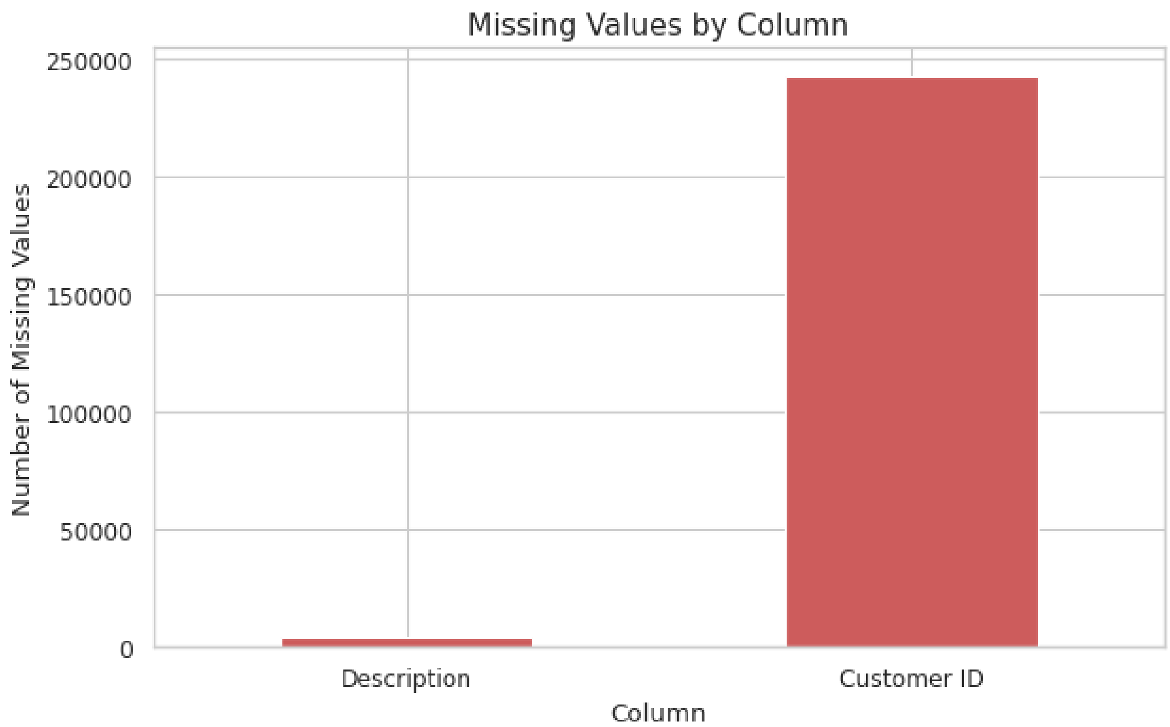

Figure 1: Missing values by column

Step 4: Top 10 Best-Selling Products

A Revenue column was created (Quantity × Price), then products were ranked by total quantity sold and by total revenue generated. The two rankings differ, which is an important business finding covered in the Insights section.

```
df['Revenue'] = df['Quantity'] * df['Price']
top_quantity = df.groupby('Description')['Quantity'].sum().nlargest(10)
top_revenue = df.groupby('Description')['Revenue'].sum().nlargest(10)
```

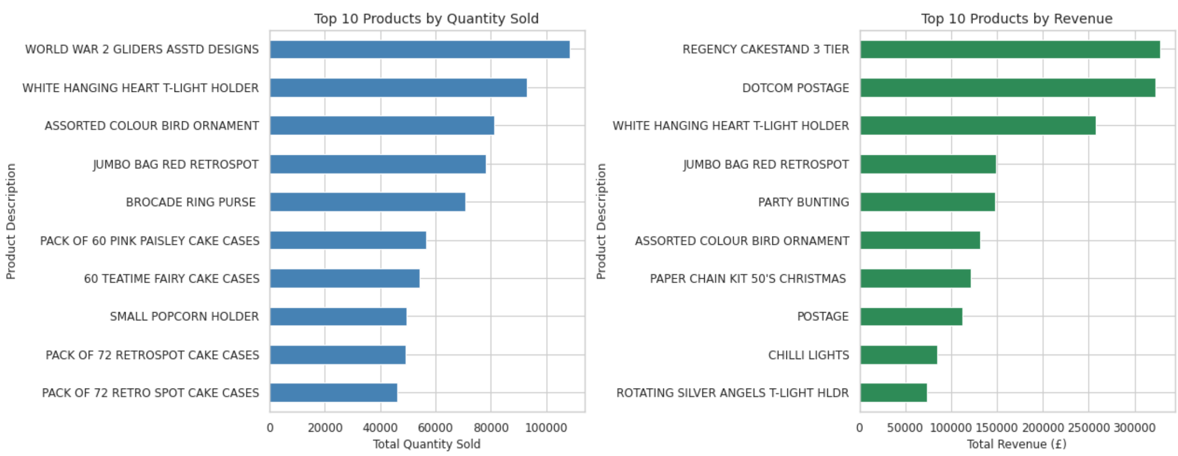

Figure 2: Top 10 products by quantity sold (left) and by revenue (right)

Step 5: Sales Performance by Country

Total revenue was aggregated by country and sorted in descending order. Because the United Kingdom dominates the dataset, a second chart excluding the UK was produced to make the remaining country trends visible.

```
Top 10 Countries by Revenue:
  United Kingdom  £16,382,580
  EIRE          £615,520
  Netherlands     £548,525
  Germany        £417,989
  France         £328,192
  Australia      £167,129
  Switzerland     £99,729
  Spain          £91,859
  Sweden          £87,809
  Denmark        £65,741
```

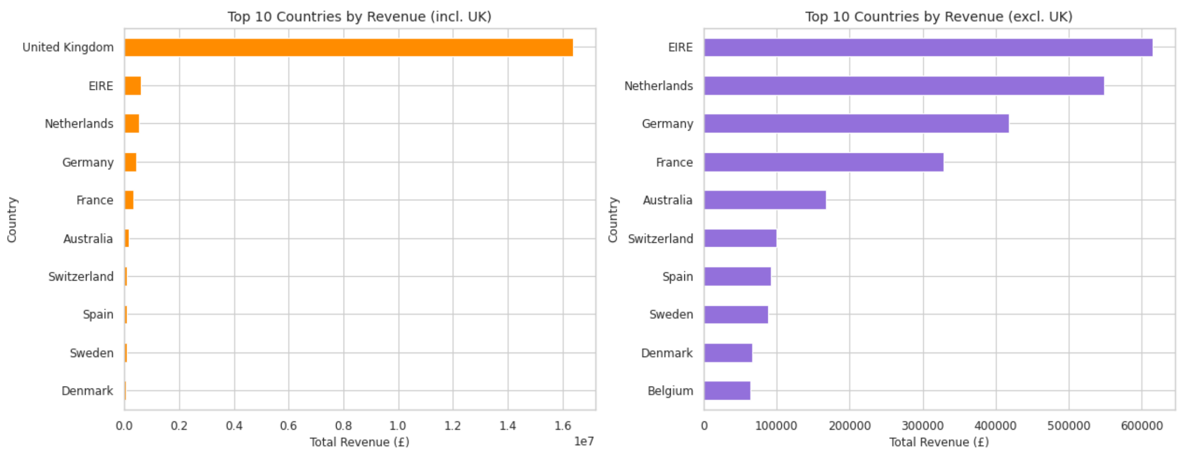

Figure 3: Top 10 countries by revenue — including UK (left) and excluding UK (right)

Step 6: Revenue Over Time (Monthly Trend)

The InvoiceDate column was converted to datetime and revenue was resampled to monthly totals to reveal seasonal trends over the two-year period.

```
Df['InvoiceDate'] = pd.to_datetime(df['InvoiceDate'])
monthly_revenue = df.set_index('InvoiceDate').resample('ME')['Revenue'].sum()
```

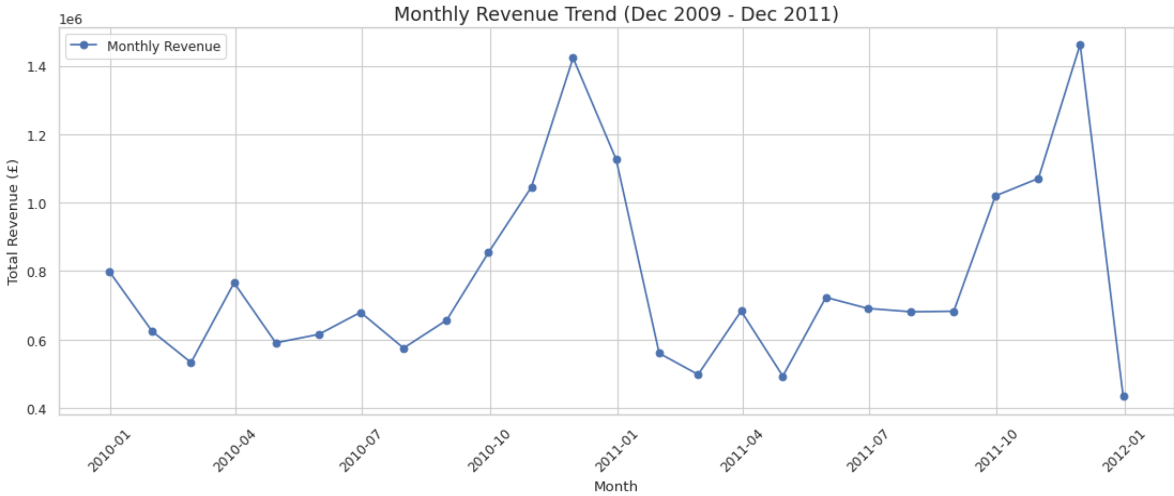

Figure 4: Monthly revenue trend, December 2009 to December 2011

Step 7: Correlation Heatmap

A correlation matrix was computed for the three numeric variables — Quantity, Price, and Revenue — and visualized as a heatmap to understand how strongly each variable relates to the others.

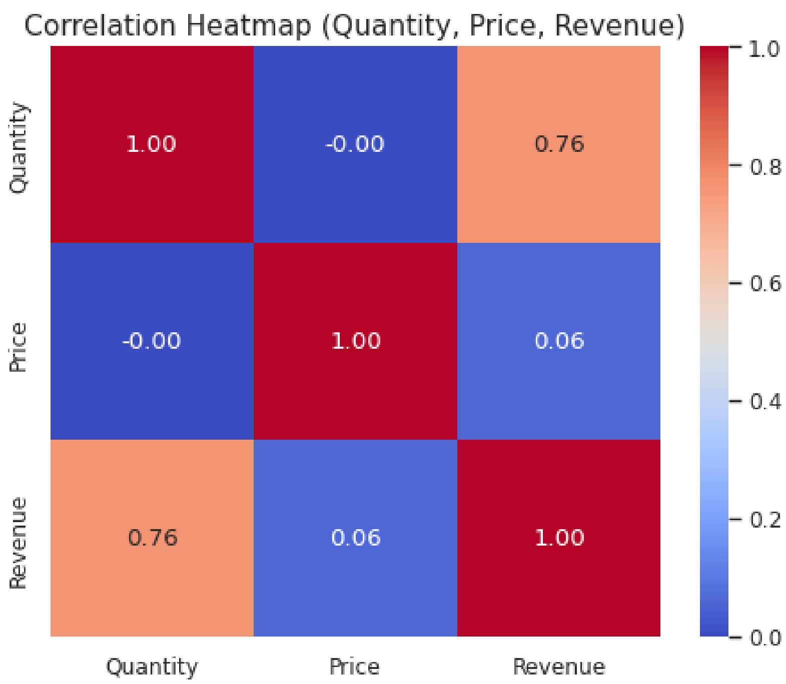

Figure 5: Correlation heatmap of Quantity, Price, and Revenue

Step 8: Outlier Detection (Box Plots)

Box plots were generated for Quantity, Price, and Revenue to visually detect outliers. As instructed, these outliers were only observed and reported — no values were removed or capped.

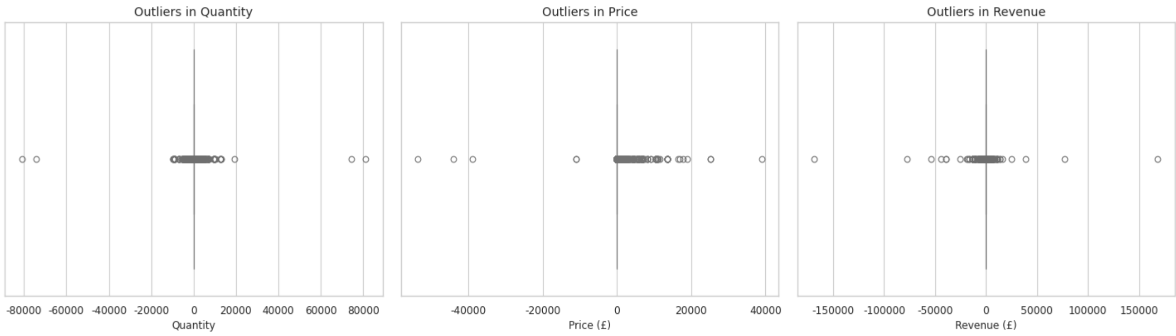

Figure 6: Box plots showing outliers in Quantity, Price, and Revenue

3. Business Insights

The following key figures were computed from the combined dataset:

Total Revenue: £19,287,250.57
Unique Customers: 5,942
Countries Served: 43
UK Share of Revenue: 84.9%
Orders with cancellations: 11,684 out of 53,628 (21.8%)

1. UK dominates revenue overwhelmingly. The United Kingdom alone contributes roughly 85% of total revenue, with EIRE, the Netherlands, and Germany forming a distant second tier. Any recommendation system should be trained primarily on UK behavior, with country as a strong feature for international customers.

2. Best sellers by quantity and by revenue are different products. Items like "WORLD WAR 2 GLIDERS ASSTD DESIGNS" and "WHITE HANGING HEART T-LIGHT HOLDER" sell in huge volumes at low unit prices, while products like "REGENCY CAKESTAND 3 TIER" and "DOTCOM POSTAGE" generate high revenue despite lower unit sales. Popularity and profitability need to be modeled separately.

3. Revenue shows clear seasonality. Monthly revenue rises sharply from October to December each year and drops off in January/February, indicating strong gift-buying seasonality that should inform inventory planning and marketing timing.

4. Data quality issues need handling before modeling. About 23,000 rows have negative Quantity (cancellations/returns), roughly 243,000 rows are missing Customer ID (guest/unidentified transactions), and there are over 34,000 duplicate rows — all of which must be addressed before building a recommendation system, since missing Customer IDs make those transactions unusable for personalization.

5. Quantity and Price outliers are common and likely reflect legitimate wholesale activity. Since this retailer sells gift and party items in bulk, extreme values in the box plots likely represent genuine wholesale purchases rather than pure data errors and should be investigated rather than blindly removed.

6. Revenue is only weakly correlated with either Quantity or Price alone. The correlation heatmap shows Revenue is driven by the combination of Quantity and Price rather than either variable individually, confirming that univariate pricing rules would not capture revenue behavior well.

7. A relatively small customer base drives all transactions. With under 6,000 unique customers generating over a million transaction line items, customer-level purchase history is rich enough to support collaborative-filtering-based recommendations.

4. Conclusion

This exploratory analysis of the Online Retail II dataset provided a clear picture of the retailer's customer base, product performance, and revenue patterns across two years of transactions. The dataset is UK-centric, seasonally driven, and contains data quality issues (missing Customer IDs, cancellations, duplicates) that must be resolved before any recommendation system is built. All findings above were produced without removing any data, in line with the task's observation-only requirement, and provided a solid foundation for the company's next phase of building a personalized recommendation system.

5. Deliverables

- Jupyter/Colab Notebook (.ipynb)full code, comments, and outputs
- This documentation (MS Word) step-by-step explanation with screenshots
- GitHub Repository code and documentation uploaded
- LinkedIn Post, project summary shared publicly